

IVANA BARNES

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EDUCATION

University of Michigan | Ann Arbor, MI | *August 2026*

M.S. in Ecology and Evolutionary Biology | **GPA: 4.0**

- Outstanding Graduate Student Instructor Award

Yale University | New Haven, CT | *May 2023*

B.S. in Ecology and Evolutionary Biology | **GPA: 3.98; summa cum laude**

- Edgar J. Boell Prize for Excellence in Biology

SKILLS

- **Programming:** R; Python (basic); Bash/Unix shell; Stan; Shiny; HPC
- **Bioinformatics:** ddRADseq; ipyrad; IQ-TREE; molecular phylogenetics
- **Evolutionary Simulation:** SLiM; tskit; msprime

RESEARCH EXPERIENCE

Graduate Researcher | *University of Michigan* | Aug 2023 – Aug 2026

- Developed an update to BEDASSLE, a Bayesian method implemented in R for analyzing isolation by environment in population genetic datasets
 - Updated the underlying likelihood function to improve scalability to larger datasets, implemented a model selection component, and moved the MCMC to RStan
 - Simulated >1,000 genomic datasets to validate the updated method, using SLiM, tskit, and msprime to simulate evolution and convert tree-sequence outputs into genomic datasets
 - Designed and refined R graphics to visualize BEDASSLE model performance on simulated data
- Presented a poster on BEDASSLE at Midwest Population Genetics 2026
- Led discussion and lab sections as a teaching assistant, including R labs for both Statistical Model Building and graduate-level Evolution
- Attended the 2024 MBL Workshop on Molecular Evolution at Woods Hole

Undergraduate Research Assistant | *Yale University* | May 2021 – May 2023

- Prepared mtDNA and ddRADseq libraries from >300 crayfish specimens, processed sequencing data using ipyrad, and inferred molecular phylogenies using IQ-TREE
- Presented work as a talk at the Yale EEB Senior Symposium and as a conference poster

PUBLICATIONS

- PNAS, 2025 — co-author, conservation impacts of mosaic hybrid zones
- BioInvasions Records, 2023 — co-author, identification of a nonnative crayfish species
- American Journal of Human Biology, 2022 — co-author, relationship between media and body image among the Qom, an Indigenous population in Argentina